

Chapter 4

Preliminary Results

The preliminary results use the lowest-fidelity 0D scaled model to generate improved DNP group parameters. The results verify the DNP group parameters using stationary thermal fission of ^{235}U due to the abundance of data in the literature for this case. These preliminary results investigate individual DNPs, DNP group parameter sensitivities, and DNP group parameter verification.

Of the MSR physical phenomena considered, only chemical removal has an impact on the 0D scaled model. In-core and ex-core residence times do not change delayed neutron yield per fission because the number of fission events producing that count rate also varies. In the simplified 0D scaled model, each DNP source scales linearly at the same rate as the number of fissions. The proposed 0D flow model will not have this linear response, as it accounts for time-dependent decays and parasitic absorptions.

Specifically, for the scaled flux method applied to the cumulative fission yield, the scaling term for both is the same. Although this approach is rapid and enables more simulations, it does not incorporate fuel salt circulation into improved DNP group parameter generation. Approximate measured in-core and ex-core residence times for the MSRE are 9 seconds and 16 seconds, respectively [71].

An additional consideration is that the pulse irradiation results are not included. Whether combined via simultaneous solution of a larger matrix or solved separately and then combined, pulse irradiation results improve DNP group parameters for shorter-lived groups [89]. Longer-lived groups are expected to be more accurate than shorter-lived groups in the 0D scaled model, as it excludes pulse irradiation results. Section 5.1.1 discusses this further.

These results use the parameters in Table 4.1, unless otherwise specified. The sampled simulations row shows the number of non-linear least squares solves generating improved DNP group parameters. For each sampled simulation, emission probability, half-lives, and fission yields are sampled within uncertainties using a normal distribution.

4.1 Important DNPs

Because the microscopic approach uses individual DNPs, it is possible to analyze the contribution of each DNP to the total delayed neutron yield. This is given in Equation (2.28), recreated here as:

$$\bar{\nu}_d = \sum_{i=1}^I P_{n,i} CFY_i.$$

Table 4.1: Default parameters for 0D scaled model simulations.

Parameter	Value
Fissile Nuclide	^{235}U [-]
Incident Neutron Energy	0.0253 [eV]
Fission Yields	ENDF/B-VII.1 [34]
Half-lives	IAEA [50]
Emission Probability	IAEA [50]
Final Decay Time	1200 [s]
Decay Time Spacing	Logarithmic
Decay Time Nodes	300
Sampled Simulations	1000

Another quantity of interest is the number of delayed neutrons from the existing concentration of nuclides, rather than the number of delayed neutrons per fission. This can be calculated by integrating the delayed neutron count rate, as shown in Equation (2.27), recreated here as:

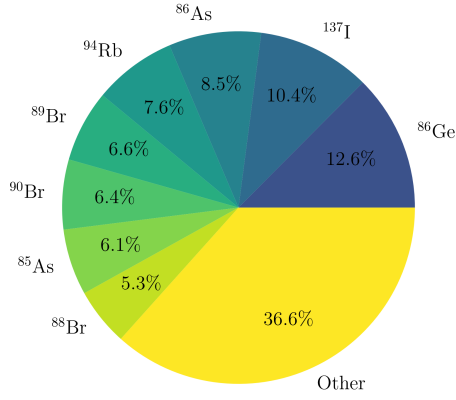
$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0).$$

Figure 4.1 shows both approaches and how they differ. Figure 4.2(a) shows the delayed neutron yield contributions, which indicate the DNPs which emit the most delayed neutrons after a single fission event. Figure 4.2(b) shows each DNPs' contribution to the delayed neutron count rate for a saturation irradiation. The count rate contribution is weighed more heavily towards longer-lived DNPs because they build up during the saturation irradiation. Figure 4.2(c) shows the concentrations of each DNP, correlated with Figure 4.2(b). DNPs with a large concentration may have a small emission probability and contribute few delayed neutrons to the count rate, such as ^{141}Cs .

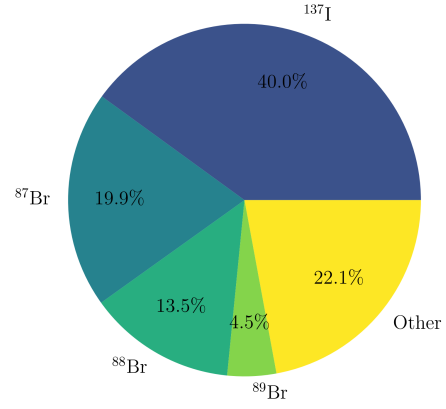
For the delayed neutron yield contributions in Figure 4.2(a), ^{86}Ge contributes the most, meaning that for each thermal fission event in ^{235}U , approximately 12.5% of delayed neutrons come from ^{86}Ge . ^{86}Ge is exceedingly short-lived, with a half-life of approximately 200 milliseconds. In Figure 4.2(b), it contributes less than 4% of the total delayed neutrons after the sample has been irradiated. Longer-lived nuclides have accumulated during irradiation, dominating the delayed neutron count rate. This reinforces why pulse irradiation data are important, as saturation irradiation does not fully capture short-lived DNPs effects. Figure 4.2(c) shows the relative concentration of each DNP. The concentrations are generally consistent with the literature, though ^{141}Cs is an outlier [99]. This may result from the 0D scaled model simplifications, such as excluding parasitic absorption, or because other works ignore ^{141}Cs due to its extremely small emission probability.

Figure 4.2 shows the delayed neutron count rate in Figure 4.2(a), as well as the integrated form in Figures 4.2(b) and 4.2(c), representing the total delayed neutron counts at each time. Figure 4.2(a) shows the ^{86}Ge count rate dominates at short time scales due to its short half-life, but it becomes less important after a tenth of a second. This again demonstrates how saturation irradiation enables better analysis of the longer-lived DNPs, but reduces the relative importance of shorter-lived DNPs. Figure 4.2(b) shows total counts varying over time on a log scale. ^{86}Ge initially dominates but quickly becomes unimportant compared to other DNPs. Figure 4.2(c) mirrors Figure 4.1(b), as both show ^{86}Ge contribution is minimal after saturation irradiation. The total delayed neutron count contribution is dominated by longer lived DNPs like ^{137}I and ^{87}Br .

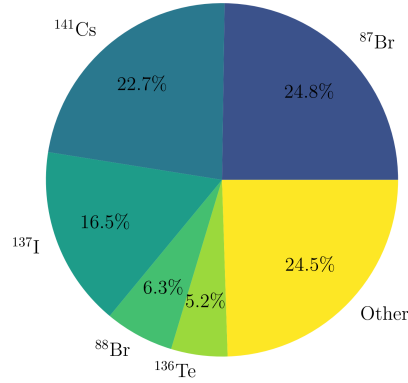
Another analysis of interest with the individual DNP data is calculation of the delayed neutron yield and average half-life, shown in Table 4.2. These values are calculated and then compared with the six group DNP group parameters formulated from those DNPs. Ideally, these values would be identical, but the six-group fit cannot fully capture all decaying DNPs behavior, producing small differences. These differences are small relative to the



(a) Delayed neutron yield contributions.



(b) Delayed neutron count rate contributions.



(c) DNP relative concentrations.

Figure 4.1: Delayed neutron contributions from each DNP after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

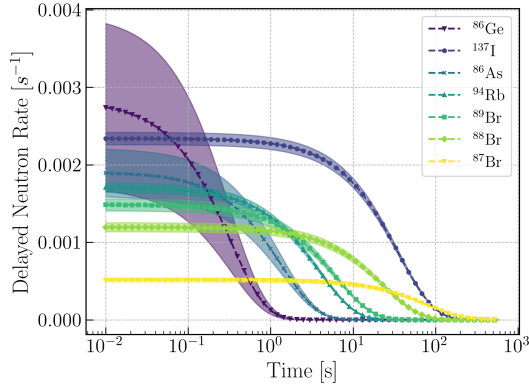
uncertainty bounds for each term.

Table 4.2: The delayed neutron yield and average half-life calculated by summing individual DNPs and from the 6 DNP group parameters.

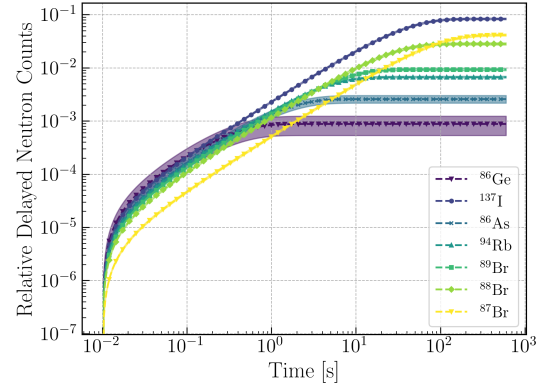
Term	DNP	Group
$\bar{\nu}_d$	0.0225 ± 0.0015	0.0224 ± 0.0013
$\bar{\tau}[\text{s}]$	6.3852 ± 0.3801	6.4029 ± 0.3365

4.2 Sensitivity Studies

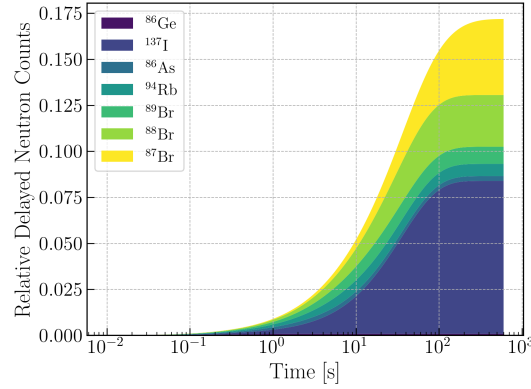
This section examines DNP group parameter sensitivity to various simulation parameters. The evaluated simulation parameters are chemical removal, time nodes for the non-linear least squares problem, and nuclear data uncertainties. For the chemical removal analysis, the MSBR chemical reprocessing scheme is used. The MSBR reprocessing scheme has large chemical removal rates, providing a conservative impact of these chemical removal rates on the DNP group parameters. The time node evaluation includes the spacing between each node, the number of nodes,



(a) The delayed neutron count rate.



(b) The total delayed neutron counts.



(c) A stack plot of the total delayed neutron counts.

Figure 4.2: Delayed neutron count rate contributions from the top two most contributing DNPs at each time step after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

the total amount of time the sample should decay, and a combination of the total time and number of nodes. Finally, the nuclear data uncertainties include uncertainty in the fission yields, half-lives, and emission probabilities. To propagate these uncertainties to DNP group parameters, the non-linear least squares solver runs 1000 times using random normal samples of each DNP's fission yield, half-life, and emission probability.

4.2.1 Chemical Removal

As shown in Appendix B, the fraction of fuel salt in the ex-core region determines the scaling of the chemical removal rate. This analysis assumes an ex-core fraction of 0.5. This value is based on the core zone I, core zone II, and annulus salt volumes relative to primary system salt volume in the MSBR [148]. Table 3.1 shows the MSBR chemical reprocessing scheme used in this analysis, recreated here.

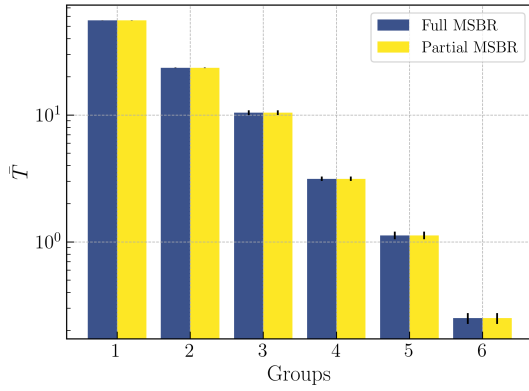
Table 4.3: MSBR Online Reprocessing Removal Rates from [148].

Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

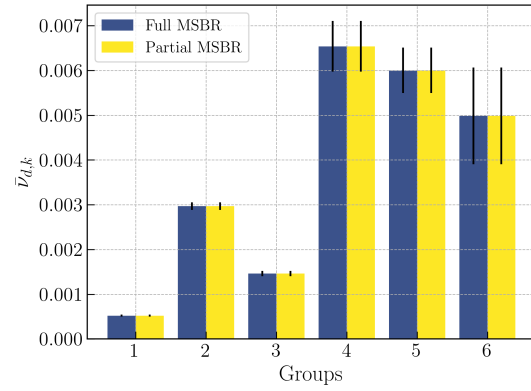
One consideration is that the 0D scaled model underestimates chemical removal rate effects. This is because it only chemically removes from the cumulative fission yield of the element in the reprocessing scheme, and does not account for decay chains which may have chemical removal rates included. For example, ^{137}I has a cycle time of 60 days, corresponding to a chemical removal rate of $1.9 \times 10^{-7} [s^{-1}]$. However, Sb and Te, which decay into I, have a chemical removal rate of $0.05 [s^{-1}]$, many orders of magnitude larger than the I removal rate. The 0D scaled model, with cumulative fission yields and no decay chain, does not capture this impact on the concentration of iodine.

Long Cycle Time Evaluation

Table 3.1 lists many elements to consider. Since the sample is irradiated for only several minutes, it is useful to assess the significance of smaller removal rates, such as the 500-day cycle time of rare earths. This analysis considers only 20-second cycle time elements, the noble metals and gases. Figure 4.3 shows the difference between the full MSBR reprocessing scheme and the partial scheme considering only 20-second cycle time elements. The differences between each DNP group parameter, $\bar{v}_{d,k}$ and λ_k , are less than 0.001%. This analysis shows elements with cycle times longer than 20 seconds do not affect DNP group parameters for thermal saturation irradiation of ^{235}U using the 0D scaled model.



(a) Half-life DNP group parameters.

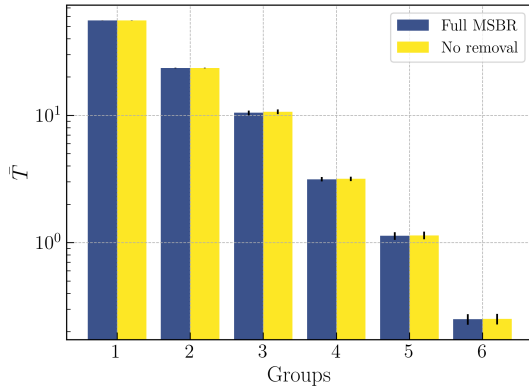


(b) Yield DNP group parameters.

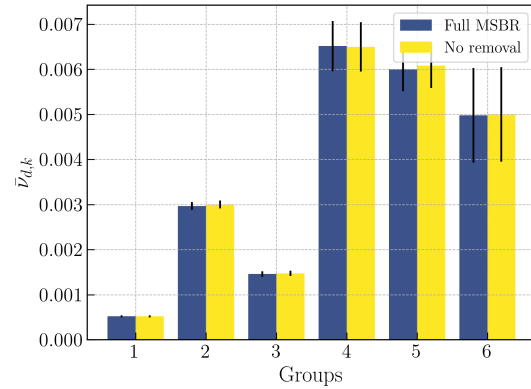
Figure 4.3: Delayed neutron contributions from each DNP with and without long cycle time chemical removal rates in the MSBR after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Removal Rate Evaluation

It is also of interest to determine the sensitivity of the DNP group parameters to the chemical reprocessing scheme. The 0D scaled model underestimates chemical removal effects due to the use of cumulative fission yields without explicit nuclide tracking, but this analysis still demonstrates the general chemical reprocessing impact. The comparison of DNP group parameters with and without the MSBR reprocessing scheme is given in Figure 4.4. The figure shows the half-lives are similar, while yields without removal are slightly higher.



(a) Half-life DNP group parameters.



(b) Yield DNP group parameters.

Figure 4.4: Delayed neutron contributions from each DNP with and without the MSBR chemical removal rates after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Differences between average half-lives are negligible, while differences between the total delayed neutron yields are 0.57%, or approximately 12 pcm. This small difference likely results from the 0D scaled model underestimating the effect of chemical removal. Since chemical reprocessing consistently reduces total delayed neutron yield by removing DNPs, it is important to account for this factor.

4.2.2 Time Nodes

The time node sensitivity is important, as many nodes may be required to fully capture the physics, but each additional node increases computational cost. Of interest are the node distribution, number of nodes, sample decay time, and combined decay time and number of nodes.

Distribution

Two time node distributions are compared: logarithmic spacing from 0.01 to 600 seconds and linear spacing from 0 to 600 seconds. With 300 time nodes, linear spacing does not capture behavior in the first 2 seconds but provides higher resolution at longer time scales. This longer time scale resolution is less useful, as many relevant DNPs quickly drop off at longer time scales, as discussed in Section 4.1.

Figure 4.5 compares DNP group parameters for logarithmic and linear spacing. Half-lives of shorter-lived groups are larger for linear spacing, consistent with expectations. Linear spacing has no data between zero and two seconds, and it places greater weight on longer-lived groups due to the data distribution. This causes shorter-lived groups to capture a larger yield, with yields shifted from groups two, three, and four to groups five and six. The total DNP yield differs by approximately 0.1%, so changing time node spacing does not alter macroscopic DNP group behavior. Instead, it affects the distribution of delayed neutrons over time, with logarithmic spacing more accurate by capturing shorter-lived DNPs effects.

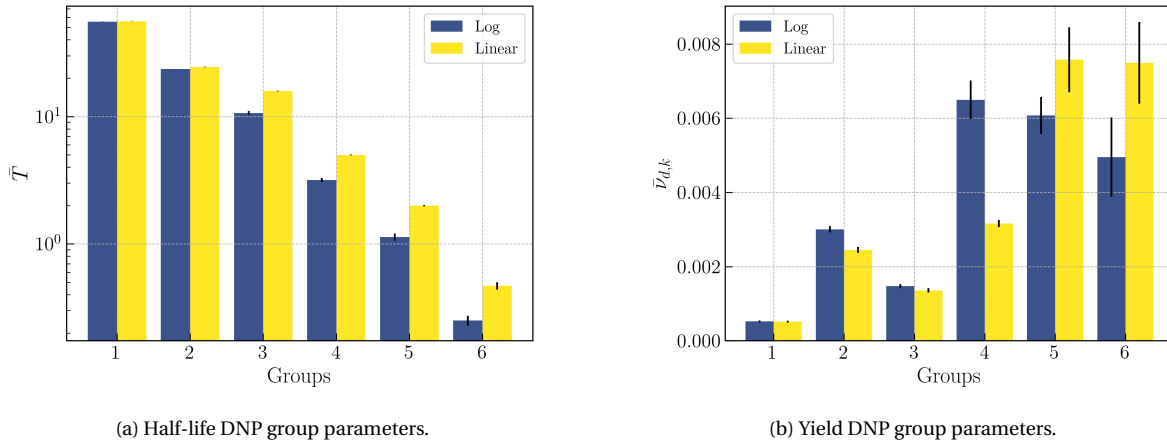


Figure 4.5: DNP group parameters for linear and logarithmic spacings of the time nodes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Number of Nodes

Figure 4.6 shows how varying the number of nodes affects the results. It appears that there is no significant difference between the results when varying the number of nodes. The main difference appears in the yields of groups four and five, with increasing nodes increasing the yield of group four and decreasing the yield of group five. There are also some smaller differences, such as the yield of group six decreasing and the yields of groups two and three increasing with an increasing number of nodes.

The simulation with the most nodes is most accurate, which allows calculation of percent differences in DNP group yields and the associated computational cost. This is shown in Table 4.4, as 50 nodes results in the group five

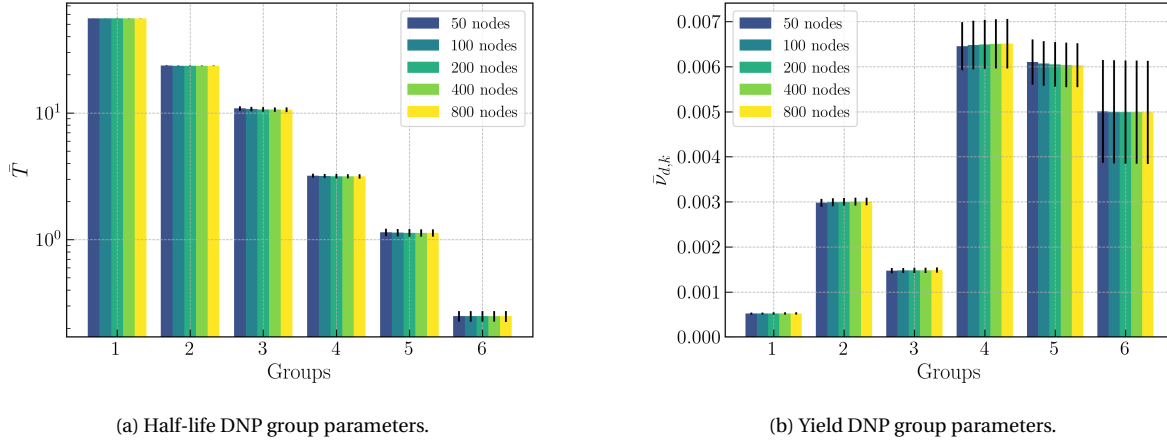


Figure 4.6: DNP group parameters for varying numbers of time nodes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

yield being off by approximately 1%, and 200 nodes keeps differences below 0.02%. Computational cost remains fairly constant across these node numbers. These results indicate that 300 nodes provide sufficiently accurate yields, within 0.1%, while remaining in the constant computational cost regime.

Table 4.4: Comparison of how the number of nodes changes the computational cost and relative difference in yields compared to the 400 node case after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Nodes	Yield Difference [%]						Cost [s]
	1	2	3	4	5	6	
50	0.30	0.90	0.53	0.75	0.99	0.59	297
100	0.17	0.49	0.25	0.39	0.46	0.77	297
200	0.01	0.02	0.00	0.01	0.01	0.09	300
400	0.00	0.00	0.00	0.00	0.00	0.01	304
800	-	-	-	-	-	-	310

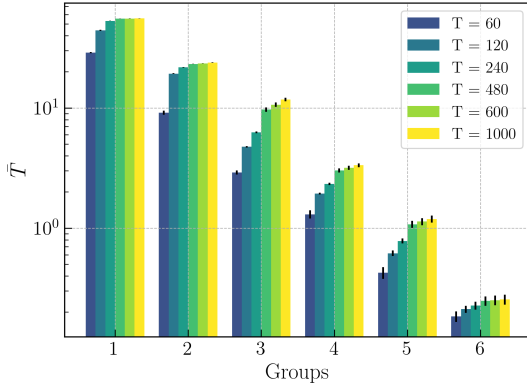
Total Decay Time

Increasing total decay time produces two simultaneous effects: decreasing time node density, discussed in Section 4.2.2, and increasing the number of nodes evaluated at longer times. Because the longest-lived DNP¹, ^{87}Br , has a half-life of approximately one minute, it is reasonable to assume that the default final time of ten minutes offers sufficient data on all DNP decays.

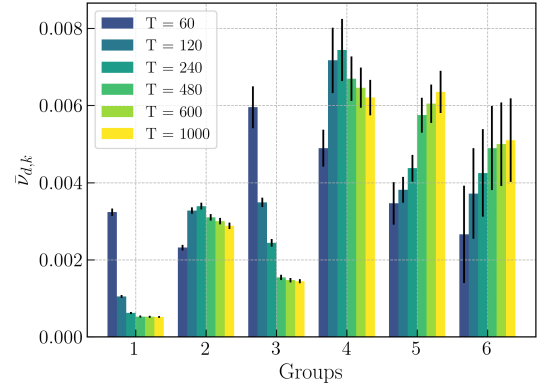
Figure 4.7 shows this assumption is not accurate. The DNP group parameters approach asymptotic values as the total decay time increases, with the decay time of 1200 seconds, or 20 minutes, being the smallest decay time after which longer decay times do not change the results. The shorter total decay times inaccurately model DNPs, as they do not fully capture long-lived DNPs and have shorter group half-lives. These shorter half-lives correlate with an overestimation of some longer-lived DNP group yields.

The total delayed neutron yields and average half-lives, shown in Table 4.5, provide another comparison between different decay times. The table shows differences are small and within uncertainty bounds.

¹²¹I, ²¹⁰Tl, and ²¹⁰Hg are longer-lived and have non-zero emission probabilities, but they live only tens of seconds longer and generally have negligible concentrations.



(a) Half-life DNP group parameters.



(b) Yield DNP group parameters.

Figure 4.7: DNP group parameters for varying total decay times after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

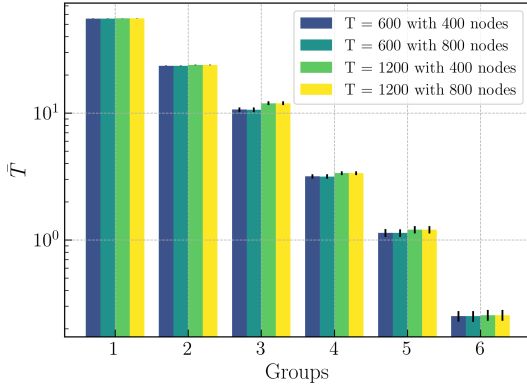
Table 4.5: Comparison of how the net decay time, T changes the total delayed neutron yield and average half-life after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

T [s]	$\bar{v}_d$	$\bar{\tau}$ [s]
150	0.0225 ± 0.0015	6.3786 ± 0.3784
300	0.0225 ± 0.0014	6.3945 ± 0.3554
600	0.0225 ± 0.0013	6.3978 ± 0.3412
1200	0.0225 ± 0.0013	6.3977 ± 0.3415
2400	0.0225 ± 0.0013	6.3977 ± 0.3417
4800	0.0225 ± 0.0013	6.3977 ± 0.3417

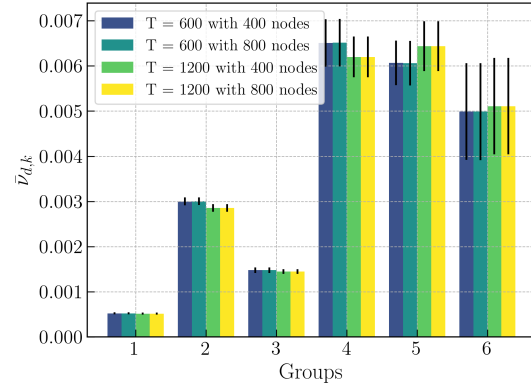
Time Node Density

Section 4.2.2 shows how the total decay time affects the results, and Section 4.2.2 shows how the number of time nodes affects the results. This section evaluates DNP group parameter sensitivity to the node density, number of nodes, and decay time by comparing different numbers of nodes and decay times. Figure 4.8 shows how the DNP group parameters vary with node density.

If the DNP group parameters primarily depend on the node density, a 1200 second decay time with 800 nodes would be equivalent to a 2400 second decay time with 1600 nodes. Instead, the results remain approximately constant with varying node numbers, with the main difference arising from final decay time. This indicates the sensitivity analysis in Section 4.2.2 is valid and does not need to account for varying time node density.



(a) Half-life DNP group parameters.



(b) Yield DNP group parameters.

Figure 4.8: DNP group parameters for varying node densities after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

4.2.3 Nuclear Data Uncertainties

A Monte Carlo approach evaluates DNP group parameter sensitivity to nuclear data uncertainties. For each nuclear data variable in the 0D scaled model, such as ^{87}Br half-life, a random value is sampled from a normal distribution. This process is repeated for every DNP fission yield, half-life, and emission probability.

Count Rate

Figure 4.9 shows count rate normalized to the Keepin et al. count rate. The red lines show count rates from sampled simulations, while the blue dashed line shows the DNP group parameters and uncertainty based on 1000 Monte Carlo non-linear least squares solves. The black X's indicate the mean count rate at each point. The other six-group DNP parameters plotted include Keepin (1.0, as the plot is normalized to Keepin values), Brady, and Synetos [89, 23, 179].

Beyond approximately three seconds, the current DNP group parameters agree within uncertainty bounds with most literature results, with Brady parameters showing a noticeable difference at longer times. Short time scales are inaccurate in the 0D scaled model due to the lack of pulse irradiation data, as discussed in the chapter introduction. Adding this data is expected to improve the fit at earlier times. To demonstrate this, the first two groups are used while the shorter-lived groups three through six are replaced with the Keepin et al. DNP group parameters [89]. Figure 4.10 shows this modified set of DNP group parameters.

The modified DNP group parameters are not a perfect comparison, as the non-linear least squares solve adjusts each group's yield relative to the yields and half-lives of other groups. This indicates excluding pulse irradiation data also affects longer-lived groups, though less than it affects shorter-lived groups. Adjusting the shorter-lived groups improves agreement with the literature for the 0D scaled model, particularly at times less than one second.

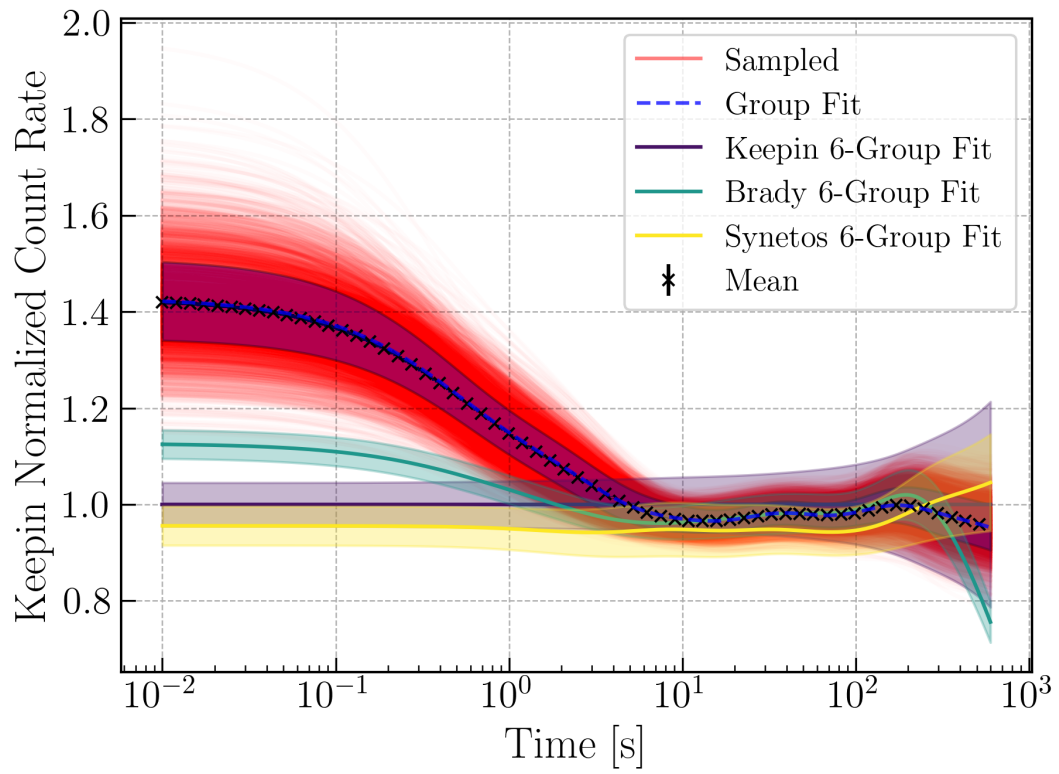


Figure 4.9: The delayed neutron count rate for thermal fission of ^{235}U normalized to the Keepin et al. six-group DNP group parameters.

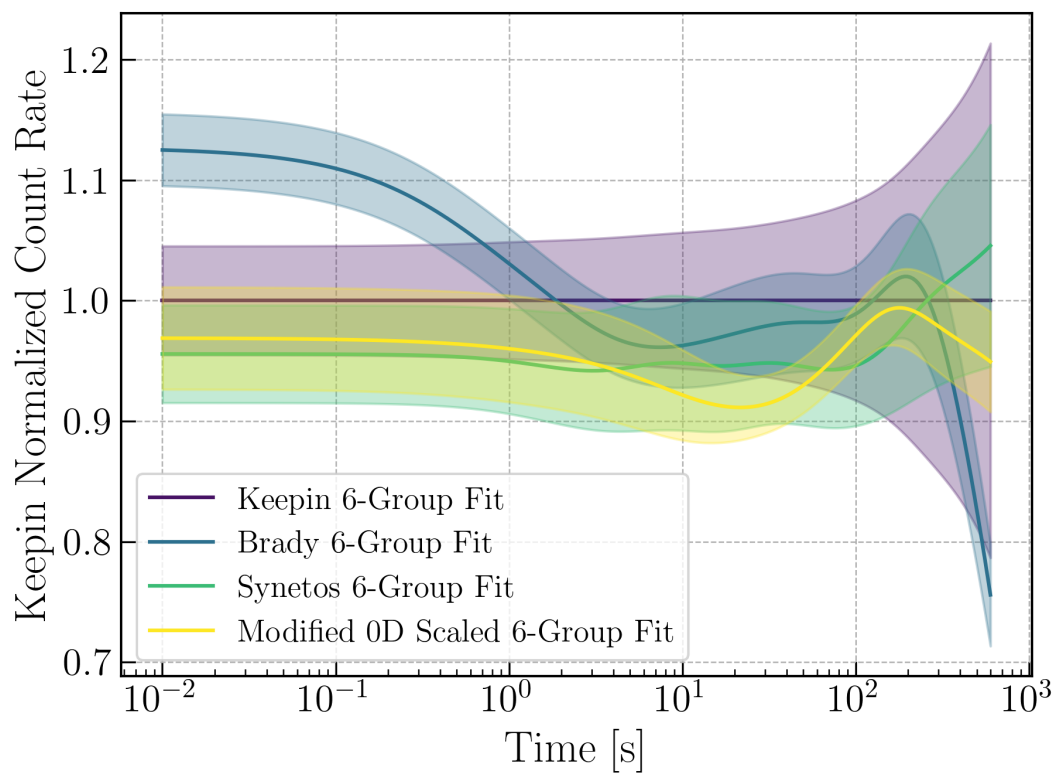


Figure 4.10: The delayed neutron count rate for thermal fission of ^{235}U normalized to the Keepin et al. six-group DNP group parameters.

Individual DNPs

The Monte Carlo sampling of each DNP's nuclear data allows for sensitivity analyses on how each change affects the DNP group parameters. For example, because group 1 is dominated by ^{87}Br , changes to its half-life, emission probability, or fission yield noticeably affect the group 1 DNP half-life and yield.

The number of Monte Carlo simulations is increased to 5000 to provide sufficient data for each DNP, ensuring differences in DNP group parameters are statistically significant. To find the nuclides with the most impactful sensitivities, a linear regression fits the slope of the relative change in DNP group parameters to the relative change in the individual DNP nuclear data, while also calculating the Pearson correlation coefficient (PCC). The PCC quantifies the linear relationship between two variables, with -1 representing perfect negative correlation, +1 perfect positive correlation, and 0 no correlation [62]. Changes in the nuclear data include emission probability P_n , fission yields (affecting concentration N), and half-lives τ .

Sensitivities of interest have absolute PCC values greater than 0.2. This threshold is somewhat arbitrary but indicates a DNP has a non-negligible correlation with a DNP group parameter. Table 4.6 presents the results of this analysis.

One of the most predictable sensitivities in Table 4.6 is ^{87}Br 's effect on group one half-life and yield, as it dominates this group. Specifically, ^{87}Br data with PCCs greater than 0.2 affect group parameters as follows: emission probability on group yield, concentration on group yield, and half-life on group half-life. Figure 4.11 shows how group one parameters vary with ^{87}Br data, as well as less strongly correlated data, such as ^{87}Br emission probability on the group three half-life. Although it may seem counterintuitive, changes to ^{87}Br emission probability affect group three half-life, showing a non-zero correlation. This is because the DNP group parameters cannot be considered in isolation; they form a collective set that best fits the delayed neutron count rate curve. Changes to that curve propagate to all group parameters, with varying magnitudes across groups and parameters.

Another DNP of interest in Table 4.6 is ^{86}Ge , an important contributor to delayed neutron yield, as shown in Figure 4.1. ^{86}Ge impacts many DNP group parameters, especially due to uncertainty in its half-life. Figures 4.12 and 4.13 show each DNP group parameter's sensitivity to variations in ^{86}Ge data.

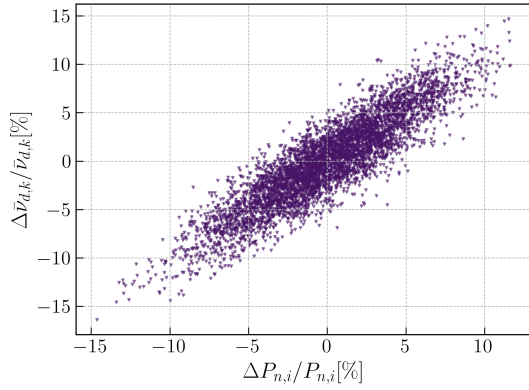
Figure 4.12 shows the largest group half-life sensitivities occur due to changes in ^{86}Ge half-life. For increasingly longer-lived groups, ^{86}Ge half-life has progressively less impact on group half-lives. This is intuitive, as a short-lived nuclide such as ^{86}Ge should have much more impact on short-lived groups than on long-lived groups. Group six half-life sensitivity to ^{86}Ge shows a lower bound with no data points, indicating asymmetry. Sensitivities typically exhibit symmetry about the line that best fits the data for a linear response. This asymmetry likely arises because many longer-lived DNPs exist, whereas ^{86}Ge is one of the few very short-lived DNPs dominating this time region. Because shorter-lived DNPs are absent in this time region, group six half-life decreases less than it increases.

Figure 4.13 shows changes in the group yields due to changes in the ^{86}Ge half-life. No clear trend exists from shorter to longer-lived groups, though groups three and four show negative correlations, suggesting a longer-lived ^{86}Ge decreases their yields while increasing their half-lives, with half-lives shown in Figure 4.12. These groups capture fewer short-lived DNPs as the ^{86}Ge half-life increases, both by increasing their group half-lives and decreasing their yields.

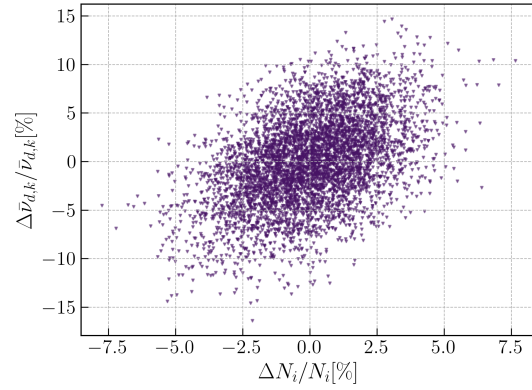
Another interesting sensitivity is the ^{86}Ge emission probability on the group six yield, showing a narrow spread for small emission probabilities that widens for larger emission probabilities. Increasing the emission probability for the short-lived ^{86}Ge corresponds to an increase in the short-lived group 6 yield. This widening likely arises from the absence of pulse irradiation data. Without this data, short-lived groups are less accurate, so removing ^{86}Ge produces a more consistent group six yield. As ^{86}Ge becomes more impactful, possible group six yield values spread due to missing pulse irradiation data.

Table 4.6: PCCs with a magnitude larger than 0.2 for variation in DNP data which correspond to changes in DNP group parameters after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

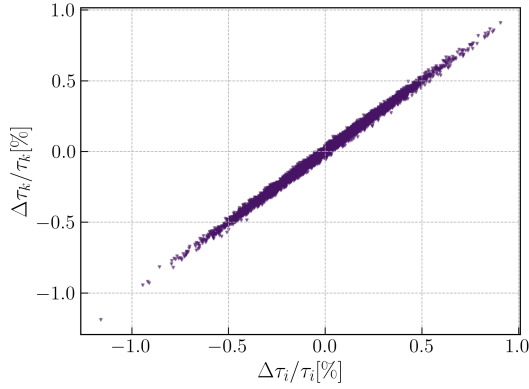
Nuclide	Group Value	DNP Value	PCC
^{84}Ga	$\tau_3[\text{s}]$	$N_i[-]$	-0.211724
^{84}Ga	$\tau_4[\text{s}]$	$N_i[-]$	-0.222001
^{84}Ga	$\tau_5[\text{s}]$	$N_i[-]$	-0.203050
^{85}As	$\bar{v}_{d,4}[-]$	$N_i[-]$	0.757084
^{85}As	$\tau_4[\text{s}]$	$N_i[-]$	-0.411166
^{85}As	$\bar{v}_{d,5}[-]$	$N_i[-]$	0.718393
^{85}As	$\tau_5[\text{s}]$	$N_i[-]$	0.448911
^{88}Br	$\bar{v}_{d,3}[-]$	$P_{n,i}[-]$	0.276294
^{88}Br	$\bar{v}_{d,3}[-]$	$\tau_i[\text{s}]$	-0.221999
^{88}Br	$\tau_3[\text{s}]$	$P_{n,i}[-]$	0.379647
^{88}Br	$\tau_3[\text{s}]$	$N_i[-]$	0.264940
^{138}I	$\bar{v}_{d,3}[-]$	$P_{n,i}[-]$	0.211254
^{137}I	$\bar{v}_{d,2}[-]$	$P_{n,i}[-]$	0.478664
^{137}I	$\bar{v}_{d,2}[-]$	$\tau_i[\text{s}]$	-0.270646
^{137}I	$\bar{v}_{d,2}[-]$	$N_i[-]$	0.746893
^{137}I	$\tau_2[\text{s}]$	$\tau_i[\text{s}]$	0.813782
^{137}I	$\tau_2[\text{s}]$	$N_i[-]$	0.252540
^{86}As	$\bar{v}_{d,5}[-]$	$N_i[-]$	0.512737
^{98}Y	$\bar{v}_{d,4}[-]$	$P_{n,i}[-]$	0.243371
^{86}Ge	$\bar{v}_{d,2}[-]$	$\tau_i[\text{s}]$	-0.238309
^{86}Ge	$\tau_2[\text{s}]$	$\tau_i[\text{s}]$	0.359997
^{86}Ge	$\bar{v}_{d,3}[-]$	$\tau_i[\text{s}]$	-0.637401
^{86}Ge	$\tau_3[\text{s}]$	$\tau_i[\text{s}]$	0.656167
^{86}Ge	$\bar{v}_{d,4}[-]$	$\tau_i[\text{s}]$	-0.407397
^{86}Ge	$\tau_4[\text{s}]$	$\tau_i[\text{s}]$	0.673363
^{86}Ge	$\bar{v}_{d,5}[-]$	$\tau_i[\text{s}]$	0.263800
^{86}Ge	$\tau_5[\text{s}]$	$\tau_i[\text{s}]$	0.662307
^{86}Ge	$\bar{v}_{d,6}[-]$	$P_{n,i}[-]$	0.830397
^{86}Ge	$\bar{v}_{d,6}[-]$	$N_i[-]$	0.464433
^{86}Ge	$\tau_6[\text{s}]$	$P_{n,i}[-]$	-0.306136
^{86}Ge	$\tau_6[\text{s}]$	$\tau_i[\text{s}]$	0.893433
^{87}Br	$\bar{v}_{d,1}[-]$	$P_{n,i}[-]$	0.896423
^{87}Br	$\bar{v}_{d,1}[-]$	$N_i[-]$	0.456307
^{87}Br	$\tau_1[\text{s}]$	$\tau_i[\text{s}]$	0.997442
^{137}Sb	$\bar{v}_{d,3}[-]$	$N_i[-]$	0.229445
^{137}Sb	$\tau_3[\text{s}]$	$N_i[-]$	-0.229578
^{137}Sb	$\tau_4[\text{s}]$	$N_i[-]$	-0.250857
^{137}Sb	$\tau_5[\text{s}]$	$N_i[-]$	-0.299691



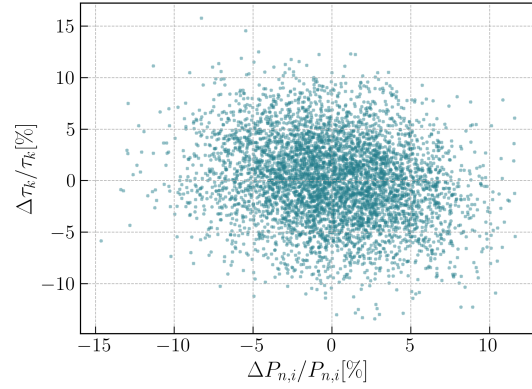
(a) DNP group 1 yield variation due to emission probability variation with a PCC of 0.90.



(b) DNP group 1 yield variation due to fission yield variation with a PCC of 0.46.



(c) DNP group 1 half-life variation due to half-life variation with a PCC of 1.00.



(d) DNP group 3 half-life variation due to emission probability variation with a PCC of -0.19.

Figure 4.11: DNP group parameter sensitivity to changes in ^{87}Br nuclear data after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

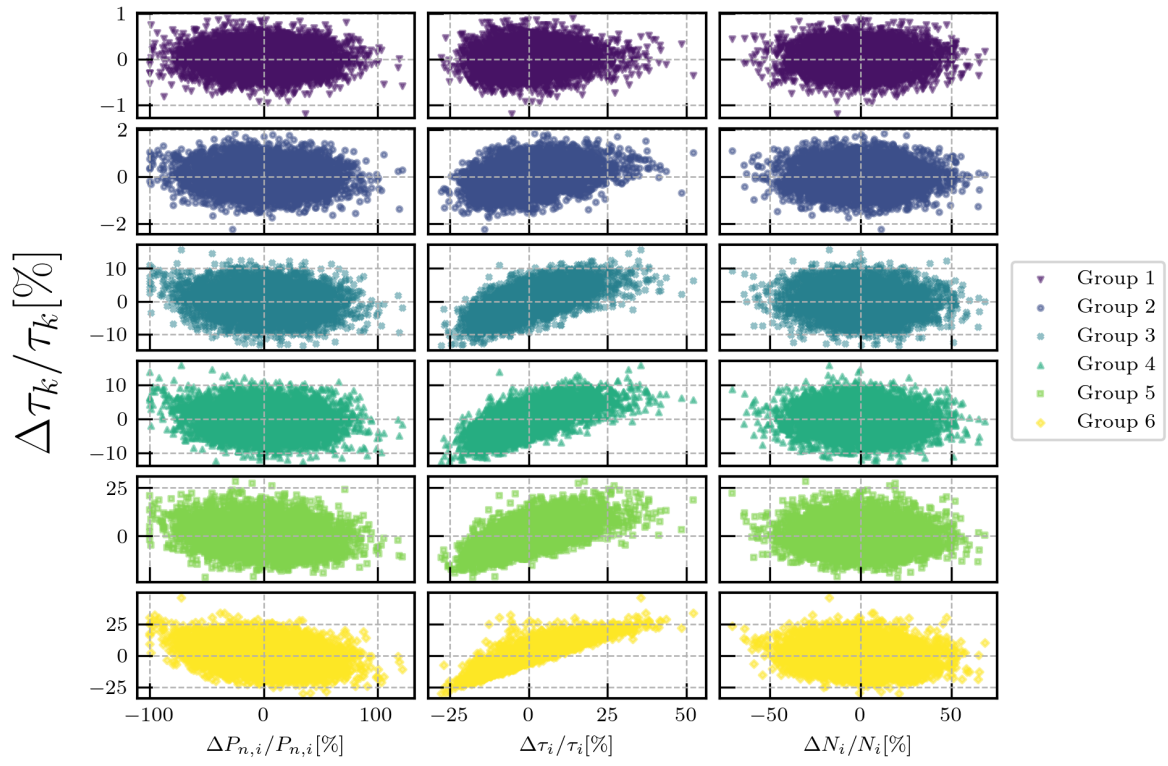


Figure 4.12: The difference in each DNP group half-life from the ^{86}Ge relative changes in emission probability, half-life, and concentration after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

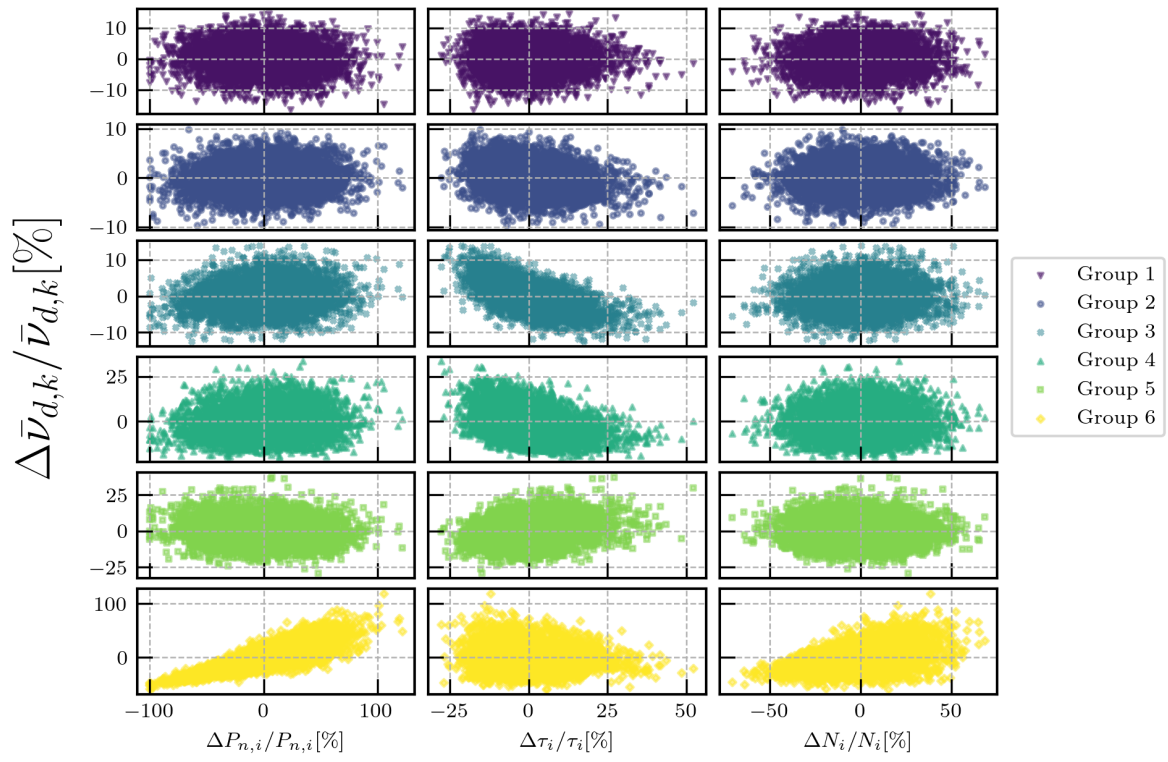


Figure 4.13: The difference in each DNP group yield from the ^{86}Ge relative changes in emission probability, half-life, and concentration after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

4.3 Verification

The 0D scaled model verification uses DNP group parameters from thermal fission of ^{235}U . This verification uses the data from literature, introduced in Section 2.1.1, combined and recreated here with this work's results.

Tables 4.7, 4.8, 4.10, and 4.10 show the DNP group parameters from literature and how they compare to the 0D scaled model DNP group parameters. As discussed, the 0D scaled model excludes pulse irradiation data, so shorter-lived groups are less accurate. This appears in the yields: groups five and six differ significantly, while groups one and two match the literature more closely. The group half-lives generally agree with the literature, though groups three and five are outliers. This likely arises from missing pulse irradiation data, as both group half-lives are approximately twice as large as the values in the literature.

Table 4.7: DNP group yields from thermal fission of ^{235}U (recreated from [128]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [89]	0.00052	0.00346	0.00310	0.00624	0.00182	0.00066
Charlton et al. [36]*	0.00054	0.00344	0.00319	0.00607	0.00186	0.00066
ENDF/B-VI.8 [153]	0.00058	0.00302	0.00288	0.00646	0.00265	0.00111
Mills et al. [119]	0.00056	0.00325	0.00196	0.00658	0.00287	0.00122
Brady and England [23]	0.00068	0.00341	0.00292	0.00611	0.00310	0.00158
Saleh et al. [160]	0.00055	0.00363	0.00297	0.00593	0.00168	0.00063
Synetos and Williams [179]*	0.00051	0.00328	0.00214	0.00583	0.00270	0.00063
Tuttle [187]	0.00065	0.00362	0.00320	0.00692	0.00218	0.00044
Waldo et al.[192]	0.00055	0.00366	0.00327	0.00660	0.00192	0.00070
0D scaled model	0.00052	0.00297	0.00146	0.00650	0.00597	0.00500

* Modified to include a sixth group by Parish et al. [128].

Table 4.8: DNP group yield uncertainties from thermal fission of ^{235}U (recreated from [128]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [89]	0.00006	0.00025	0.00041	0.00047	0.00019	0.00013
Charlton et al. [36]*	0.00003	0.00005	0.00024	0.00013	0.00014	0.00014
ENDF/B-VI.8 [153]	-	-	-	-	-	-
Mills et al. [119]	0.00004	0.00023	0.00020	0.00051	0.00035	0.00012
Brady and England [23]	0.00004	0.00019	0.00016	0.00034	0.00017	0.00009
Saleh et al. [160]	0.00005	0.00032	0.00027	0.00104	0.00015	0.00012
Synetos and Williams [179]*	0.00004	0.00020	0.00036	0.00037	0.00029	0.00012
Tuttle [187]	0.00007	0.00013	0.00041	0.00019	0.00021	0.00007
Waldo et al.[192]	0.00006	0.00041	0.00041	0.00038	0.00017	0.00009
0D scaled model	0.00002	0.00009	0.00006	0.00056	0.00051	0.00111

* Modified to include a sixth group by Parish et al. [128].

Table 4.11 compares the total delayed neutron yield for 100 fissions between the literature and the 0D scaled model. It can be seen that the data is not within uncertainty bounds of each other, and even differs between different approaches for the same dataset. Differences between microscopic and macroscopic delayed yields occur for many fissile nuclides, not only ^{235}U [113]. The 0D scaled model yield exceeds other literature values, but this is a data issue and not a fitting issue. The summed DNP data gives approximately the same yield as shown in Table 4.2.

Table 4.9: DNP group half-lives from thermal fission of ^{235}U (recreated from [128]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [89]	55.89897	22.72614	6.24457	2.30281	0.60802	0.23028
Charlton et al. [36]*	55.45177	22.43195	6.02737	2.18658	0.49159	0.23028
ENDF/B-VI.8 [153]	51.72740	21.19716	5.73797	2.28913	0.81595	0.24295
Mills et al. [119]	55.27489	22.80089	7.67860	2.77148	1.07381	0.28177
Brady and England [23]	52.11633	21.32761	5.68620	2.18727	0.70114	0.23462
Saleh et al. [160]	55.45177	22.65187	6.24457	2.31049	0.63013	0.23028
Synetos and Williams [179]*	56.35343	22.35959	7.00149	2.72893	0.77016	0.23028
Tuttle [187]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
Waldo et al.[192]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
0D scaled model	55.52977	23.54278	10.43942	3.14523	1.12586	0.25015

* Modified to include a sixth group by Parish et al. [128].

Table 4.10: DNP group half-life uncertainties from thermal fission of ^{235}U (recreated from [128]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [89]	1.35239	0.67061	0.22503	0.09181	0.08000	0.02219
Charlton et al. [36]*	0.44361	0.50817	0.10482	0.06208	0.03138	0.02219
ENDF/B-VI.8 [153]	-	-	-	-	-	-
Mills et al. [119]	0.09918	0.11723	0.37861	0.11259	0.08521	0.01020
Brady and England [23]	-	-	-	-	-	-
Saleh et al. [160]	3.99253	0.14805	0.39380	0.03851	0.01719	0.02219
Synetos and Williams [179]*	0.45816	0.28851	0.77794	0.21488	0.09413	0.02219
Tuttle [187]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
Waldo et al.[192]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
0D scaled model	0.15676	0.12560	0.44990	0.12771	0.07629	0.02325

* Modified to include a sixth group by Parish et al. [128].

This indicates no issue with DNP group parameter formulation; rather, pulse irradiation data is needed to capture smaller yields of shorter-lived DNP groups. This can be seen in Table 4.7, where the group 5 and 6 yields are greatly over-valued relative to other DNP group yields in the literature. The modified 0D scaled model, using Keepin et al. DNP group parameters for groups three through six, produces a yield more consistent with literature values.

Table 4.12 compares average half-lives between the 0D scaled model and the literature. The 0D scaled model average half-life is too short, primarily due to overvalued group five and six yields, which overweight those half-lives. The modified 0D scaled model, with groups three through six replaced by Keepin et al. DNP group parameters, aligns much more closely with literature. This again indicates the importance of pulse irradiation data to capture short-lived DNP behavior.

Table 4.11: Total delayed neutron yields of 100 fissions from thermal fission of ^{235}U (based on [81]).

Approach	Data	Delayed Yield
Microscopic	JEF-2.2 [11]	1.708 ± 0.12
Microscopic	JEF-3.1.1 [161]	1.477 ± 0.08
Microscopic	ENDF/B-VII.1 [34]	1.905 ± 0.10
Microscopic	ENDF/B-VI.8 [153]	1.670
Microscopic	Mills et al. [119]	1.645 ± 0.07
Microscopic	Brady and England [23]	1.780 ± 0.05
Microscopic	0D Scaled Model Data	2.254 ± 0.15
Microscopic	0D Scaled Model	2.256 ± 0.13
Hybrid	Modified 0D Scaled Model	1.531 ± 0.09
Macroscopic	JEF-2.2 [11]	1.655
Macroscopic	JEF-3.1.1 [161]	1.620
Macroscopic	ENDF/B-VII.0 [33]	1.585
Macroscopic	Keepin et al. [89]	1.580 ± 0.07
Macroscopic	Charlton et al. [36]	1.577 ± 0.03
Macroscopic	Saleh et al. [160]	1.645 ± 0.11
Macroscopic	Synetos and Williams [179]	1.510 ± 0.06
Macroscopic	Tuttle [187]	1.700 ± 0.05
Macroscopic	Waldo et al.[192]	1.670 ± 0.07

Table 4.12: Average half-life from thermal fission of ^{235}U .

Approach	Data	Average half-life [s]
Microscopic	ENDF/B-VI.8 [153]	7.66
Microscopic	Mills et al. [119]	8.63 ± 0.52
Microscopic	Brady and England [23]	7.90 ± 0.34
Microscopic	0D Scaled Model Data	6.39 ± 0.38
Microscopic	0D Scaled Model	6.39 ± 0.34
Hybrid	Modified 0D Scaled Model	8.74 ± 0.46
Macroscopic	Keepin et al. [89]	9.03 ± 0.63
Macroscopic	Charlton et al. [36]	8.92 ± 0.28
Macroscopic	Saleh et al. [160]	9.49 ± 0.90
Macroscopic	Synetos and Williams [179]	8.96 ± 0.56
Macroscopic	Tuttle [187]	8.84 ± 0.45
Macroscopic	Waldo et al.[192]	8.72 ± 0.73